Task – 6

What I Learned (My Understanding)

Through Milestone 3 of the **Open Deep Researcher** project, I learned how to convert a backend-only agent workflow into a **real, interactive research assistant**. Earlier, my work was mostly focused on agents and logic running in the background, but in this milestone I clearly understood how **UI, backend agents, and memory** come together to create a ChatGPT-like experience.

I learned how to:

- Build a **clean chat-based UI** using Streamlit
- Connect user input from the UI directly to my **Planner, Searcher, and Writer agents**
- Maintain **session memory**, so the assistant remembers previous messages
- Manage **new chats vs existing chats** properly
- Make the system feel more natural and conversational instead of one-time question answering

Simple Workflow (How the Model Works)

1. User types a question in the chat UI
2. UI sends the query to the backend
3. Planner Agent breaks the query into steps
4. Searcher Agent fetches relevant information
5. Writer Agent generates the final response
6. Response is sent back and displayed in the chat
7. Conversation history is stored as session memory
8. When a new chat is started, memory resets

Advantages of What I Built

- Clean and user-friendly **chat interface**
- Backend agents are **fully integrated with UI**
- Supports **continuous conversation** (not single questions)
- Easy to extend (PDF summarizer, URL summarizer, etc.)

- More realistic and practical compared to backend-only projects
- Clear separation between **UI, logic, and agents**

My Key Takeaway

This milestone made me realize that **AI is not just about models or agents**, but about how smoothly users can interact with them. Even a powerful backend is incomplete without a good interface and memory. Building this research assistant gave me confidence in creating **real-world AI applications**, not just scripts.

The available options are –

PDF Summarizer

The PDF Summarizer allows me to upload research papers or documents and quickly extract the key points. It helps in understanding long PDFs by generating concise summaries without manually reading the entire document.

URL Summarizer

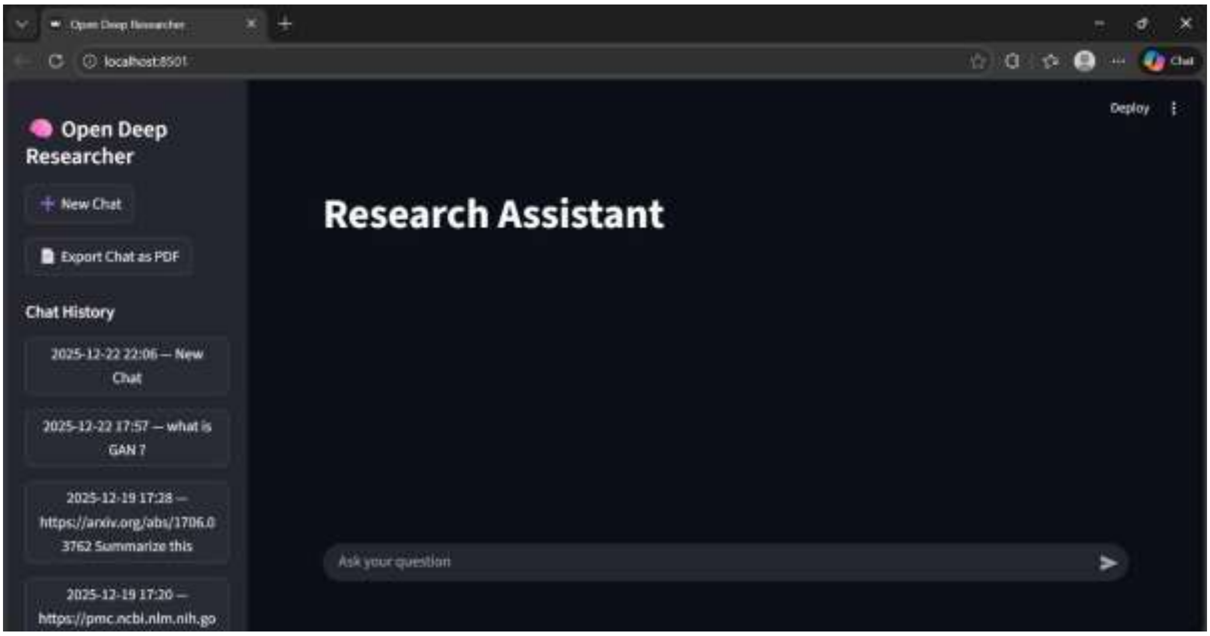
The URL Summarizer lets me provide a link to any article or research page and get a structured summary. This is useful for quickly analyzing online content and understanding its main ideas, advantages, or limitations.

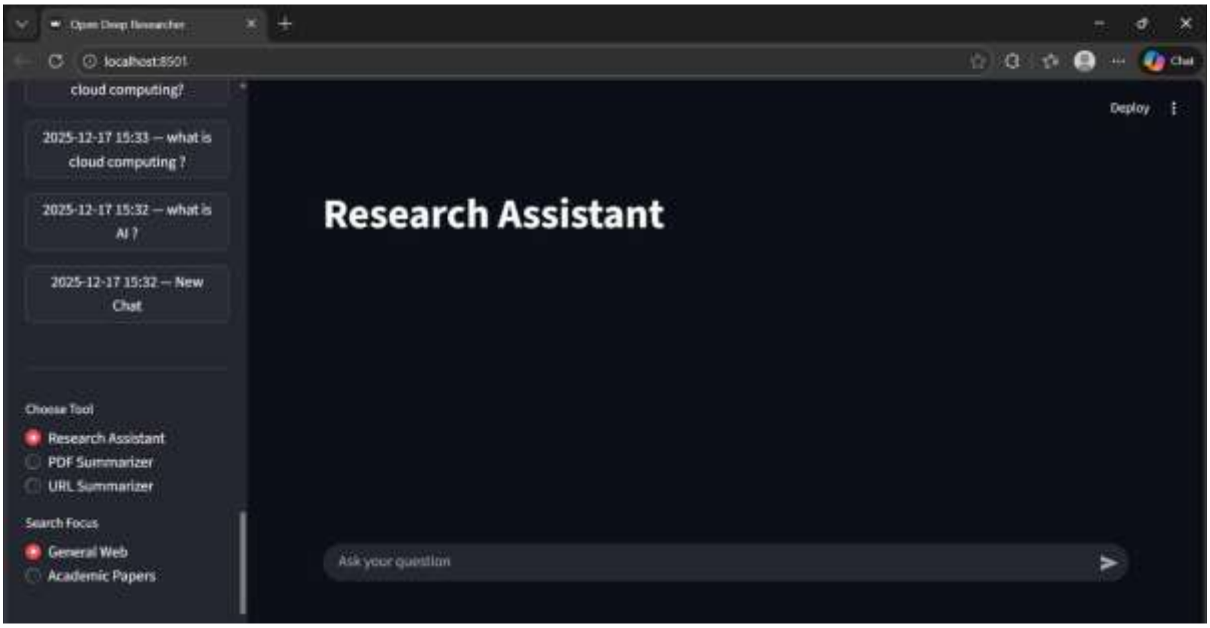
General Web (Search Focus)

When I select the General Web option, the research assistant searches across general online sources to answer my query. This is useful for broad topics, conceptual understanding, and real-world information.

Academic Papers (Search Focus)

When I choose the Academic Papers option, the assistant focuses on top research papers and scholarly sources. It analyzes academic content and provides answers that are more research-oriented and technically accurate.


localhost:8501
Deploy
Open Deep Researcher
New Chat
Export Chat as PDF
Chat History
2025-12-22 22:06 — New Chat
2025-12-22 17:57 — what is GAN ?
2025-12-19 17:28 — https://arxiv.org/abs/1706.03762 Summarize this
2025-12-19 17:20 — https://pmc.ncbi.nlm.nih.go
Research Assistant
Ask your question


localhost:8501
Deploy
cloud computing?
2025-12-17 15:33 — what is cloud computing ?
2025-12-17 15:32 — what is AI ?
2025-12-17 15:32 — New Chat
Choose Tool
Research Assistant
PDF Summarizer
URL Summarizer
Search Focus
General Web
Academic Papers
Research Assistant
Ask your question